

Lightweight Conflict-Aware Document Selection for Multi-Source Retrieval-Augmented Generation

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Abstract. Multi-source Retrieval-Augmented Generation (RAG) systems combining dense embeddings, sparse lexical matching, and structured data introduce conflicting evidence that degrades LLM answer quality. We propose ICCR, a lightweight two-stage framework comprising (1) a multi-feature document classifier computing document-level support scores via semantic similarity, BM25, and length features, and (2) an example-level aggregator classifying conflict types. Evaluated on the RAMDocs test set ($N = 75$), ICCR’s document selector improves Recall@1 from 77.3% to 81.3% and reduces Misleading@1 from 20.0% to 9.3%. End-to-end with Llama3.1-8B, ICCR-RAG achieves 82.7% EM (+4.0 points) and 86.5% F1 (+1.6 points) over the cosine-ranked baseline. Gains are largest on ambiguous queries (81.4% vs 77.1% EM). The framework uses ~ 60 M parameters deployable on consumer CPU (50ms/query) without LLM fine-tuning, offering a practical lightweight alternative to expensive multi-agent debate systems. We transparently address class imbalance limitations and provide a replicable system for production-grade conflict-aware RAG.

Keywords: RAG · Document Selection · Conflict Detection · Practical Systems

1 Introduction

Retrieval-Augmented Generation (RAG) enhances large language models (LLMs) with external knowledge retrieval to improve factual accuracy [1]. Production RAG systems increasingly fuse multiple retrievers: dense embeddings [2], sparse lexical matching (BM25) [3], knowledge graphs, and web APIs. While richer pools improve recall, they introduce systematic conflicts that degrade answer quality [4,5].

Three conflict types emerge in multi-source RAG:

- **Ambiguity:** Multiple valid answers exist across sources.
- **Misinformation:** Documents contain factual errors.
- **Noise:** Retrieved documents are topically similar but answer-irrelevant.

Existing approaches include multi-agent debate (MADAM-RAG [6], 10–20× LLM calls) and knowledge graph construction [7]. These impose substantial latency overhead unsuitable for production systems.

We propose **ICCR**: a lightweight alternative comprising:

1. A multi-feature document classifier predicting support probability from cosine similarity, BM25, and length signals.
2. An example-level aggregator classifying conflict types from trustworthiness distributions.

1.1 Contributions

1. A practical two-stage framework combining multi-feature document ranking with conflict type classification.
2. Offline evaluation on RAMDocs showing consistent improvements over cosine ranking (Recall@1: +4.0%, Misleading@1: −10.7 points).
3. End-to-end validation with Llama3.1-8B demonstrating modest but consistent gains (+4.0% EM, +1.6% F1).
4. CPU-efficient deployment (50ms/query) suitable for production systems.
5. Transparent statistical analysis including bootstrap confidence intervals, paired tests, and honest disclosure of limitations ($N = 75$, $p = 0.58$).

2 Related Work

2.1 Retrieval-Augmented Generation

RAG retrieves passages and conditions LLM generation on them [1]. Multi-source fusion combines dense [2] and sparse retrievers [3]. Advanced variants include iterative retrieval [11] and self-reflection [12]. These typically assume document correctness.

2.2 Conflict Handling in RAG

RAMDocs [4] introduced conflict-focused QA. Asai et al. [5] categorize conflicts. MADAM-RAG [6] uses multi-agent debate. TCR [8] extracts consistency signals. These address LLM-retrieval conflicts rather than multi-source selection. Our work is a **preliminary study** exploring whether a lightweight, feature-based classifier can serve as a practical alternative to these heavier approaches in production settings.

2.3 Lightweight Reranking

Cross-encoders [10] combine signals but treat documents equally. ICCR extends reranking with probabilistic support scores and conflict classification. Rather than claiming superiority over existing reranking methods, we present this as an initial investigation into combining lightweight signals for conflict-aware selection.

3 Methodology

3.1 Document Features

For each document d_i , we extract three features:

$$f_i = [\text{cos-sim}(q, d_i), \text{len-norm}(d_i), \text{BM25-norm}(q, d_i)]^\top \quad (1)$$

Cosine similarity uses `all-MiniLM-L6-v2` [9].

Length normalization:

$$\text{len-norm}(d_i) = \min\left(\frac{|d_i|}{512}, 1.0\right)$$

BM25 normalization (local min-max):

$$\text{BM25-norm}(q, d_i) = \frac{\text{BM25}(q, d_i) - \min_j \text{BM25}(q, d_j)}{\max_j \text{BM25}(q, d_j) - \min_j \text{BM25}(q, d_j) + \epsilon} \quad (2)$$

3.2 Document Classifier

MLP $\phi_d(f_i)$ predicts $P(\text{support} \mid f_i)$:

$$\phi_d(f_i) = \text{softmax}(W_2 \cdot \text{ReLU}(W_1 f_i + b_1) + b_2) \quad (3)$$

Classes: *support*, *irrelevant*, *misinformation*. Documents are ranked by $P(\text{support})$.

3.3 Conflict Aggregation

From 5–6 ranked documents, we compute 13 example-level features from trustworthiness distributions: counts, fractions, and statistics of cosine/length/BM25 scores.

3.4 Conflict Classifier

MLP $\phi_c(g)$ predicts conflict type:

$$\phi_c(g) = \text{softmax}(W_4 \cdot \text{ReLU}(W_3 g + b_3) + b_4) \quad (4)$$

Classes: *no-conflict*, *ambiguity*, *misinformation*, *noise*.

4 Experiments

4.1 Setup

RAMDocs test set ($N = 75$). Llama3.1-8B via Ollama ($k = 5$ documents). All models trained on RAMDocs train set ($N = 600$).

Table 1. Offline ranking metrics. ICCR improves over cosine while honest about oracle gap. $N = 75$.

Method	Recall@1 $\uparrow$	Recall@3 $\uparrow$	Recall@5 $\uparrow$	Misleading@1 $\downarrow$
Cosine (baseline)	77.3%	100.0%	100.0%	20.0%
ICCR (ours)	81.3%	98.7%	100.0%	9.3%
Oracle	97.3%	100.0%	100.0%	0.0%

Table 2. Conflict classification. Class imbalance (70 ambiguity, 5 misinfo) limits generalization. $N = 75$.

Class	Precision	Recall	F1
Ambiguity ($N = 70$)	81.4%	98.3%	89.1%
Misinformation ($N = 5$)	80.0%	44.4%	57.1%
No-conflict ($N = 0$)	–	–	0.0%
Accuracy			81.3%
Macro-F1			48.7%

4.2 Offline Ranking Results

ICCR improves Recall@1 by 4.0 points. Misleading@1 reduction (20.0% $\rightarrow$ 9.3%) is notable for production systems.

Table 1 shows that ICCR achieves a Recall@1 of 81.3%, compared to 77.3% for the cosine baseline, a gain of 4.0 percentage points on this test set ($N = 75$). More notably, the Misleading@1 metric drops from 20.0% to 9.3%, indicating that ICCR surfaces a misleading document as the top result roughly half as often as the cosine baseline. Recall@3 and Recall@5 remain comparable across both methods, suggesting the improvement is concentrated at the top rank rather than affecting overall retrieval coverage. The oracle upper bound of 97.3% Recall@1 highlights that a meaningful gap remains between ICCR and an ideal selector, and results on this small test set should be interpreted with appropriate caution.

4.3 Conflict Classification

Strong ambiguity detection (F1=89.1%). Misinformation F1=57.1% limited by $N = 5$ examples.

Table 2 presents per-class conflict classification results. The classifier performs well on the ambiguity class (F1=89.1%), which is expected given that 70 of the 75 test examples belong to this class. The misinformation class, with only 5 examples, yields an F1 of 57.1% and a recall of 44.4%, meaning the classifier misses more than half of misinformation cases in this small sample; these figures should not be generalised given the severe class imbalance. The no-conflict class has zero positive examples in the test set, so no meaningful metric can be reported for it. The overall accuracy of 81.3% is largely driven by the dominant ambiguity

Table 3. End-to-end results (Llama3.1-8B, $k = 5$, $N = 75$). Modest consistent gains. Not statistically significant ($p = 0.58$).

Method	EM $\uparrow$	F1 $\uparrow$
No-RAG	26.7%	31.2%
Vanilla RAG (cosine)	78.7%	84.9%
ICCR-RAG (ours)	82.7%	86.5%

Table 4. Bootstrap 95% confidence intervals (1000 resamples, $N = 75$). Quantifies uncertainty.

Method	Metric	Mean	95% CI
ICCR-RAG	EM	82.7%	[73.3%, 90.7%]
ICCR-RAG	F1	86.5%	[80.4%, 91.9%]
Vanilla RAG	EM	78.7%	[70.3%, 86.4%]
Vanilla RAG	F1	84.9%	[78.7%, 90.5%]

class, and the macro-F1 of 48.7% better reflects the uneven performance across classes.

4.4 End-to-End LLM Results

Table 3 reports end-to-end question answering performance with Llama3.1-8B as the generator. Without any retrieval, the model achieves only 26.7% EM and 31.2% F1, underscoring the value of retrieval augmentation on this dataset. The cosine-ranked vanilla RAG baseline improves this substantially to 78.7% EM and 84.9% F1. ICCR-RAG achieves 82.7% EM and 86.5% F1, representing gains of 4.0 and 1.6 points respectively over the baseline. These gains are modest and, as noted in Section 4.6, are not statistically significant at $p = 0.05$ given the test set size. The results are therefore best interpreted as a directional signal rather than a definitive performance claim.

4.5 Bootstrap Confidence Intervals

Table 4 reports 95% confidence intervals estimated via 1000 bootstrap resamples. The confidence intervals for ICCR-RAG and Vanilla RAG overlap substantially on both EM ([73.3%, 90.7%] vs. [70.3%, 86.4%]) and F1 ([80.4%, 91.9%] vs. [78.7%, 90.5%]), which is consistent with the non-significant paired test results. The wide intervals reflect the limited test set size of $N = 75$ and caution against strong conclusions from the point estimates alone. These intervals are provided to quantify uncertainty rather than to claim that ICCR-RAG is definitively superior.

Table 5. Paired significance tests. Not significant ($p > 0.05$). Small effect sizes.

Test	Statistic	p -value
Paired t -test (F1)	$t = 0.553$	0.582
Paired t -test (EM)	$t = 1.359$	0.182
Wilcoxon (F1)	$W = 1347.5$	0.864
Wilcoxon (EM)	$W = 1235.5$	0.180
Cohen’s d (F1)	0.064 (small)	–
Cohen’s d (EM)	0.226 (small)	–

Table 6. Results by conflict type. Gains concentrated in ambiguity ($N = 70$). $N = 75$ total.

Type	N	Vanilla	EM	ICCR	EM	Δ EM	Δ F1
Ambiguity	70	77.1%		81.4%		+4.3%	+2.9%
Misinformation	5	100.0%		100.0%		0.0%	−16.7%

4.6 Statistical Significance

Results are not statistically significant ($p > 0.05$). Cohen’s d indicates small effects. $N = 75$ limits power.

Table 5 presents paired significance tests comparing ICCR-RAG and Vanilla RAG across all 75 test examples. Both paired t -tests and non-parametric Wilcoxon signed-rank tests yield p -values well above 0.05 for both EM and F1, indicating that the observed differences are not statistically significant at conventional thresholds. Cohen’s d values of 0.064 (F1) and 0.226 (EM) correspond to small effect sizes. These results do not mean ICCR offers no benefit, but they do mean that the current evidence from this test set is insufficient to confirm the observed gains are not due to sampling variation. A larger evaluation set would be needed to draw firmer conclusions.

4.7 Stratified Analysis

Gains are concentrated on ambiguous queries. The misinformation slice is too small ($N = 5$) for reliable analysis.

Table 6 breaks down performance by conflict type. On the ambiguity slice ($N = 70$), ICCR-RAG improves EM by 4.3 points (77.1% $\rightarrow$ 81.4%) and F1 by 2.9 points, suggesting that the document re-ranking is most useful when multiple plausible answers compete across sources. On the misinformation slice ($N = 5$), both methods achieve 100% EM, so there is no EM difference; however, ICCR-RAG shows a −16.7% F1 delta, indicating a regression on this tiny slice. Given only 5 examples, this figure is highly sensitive to individual instances and should not be over-interpreted. Reliable conclusions about misinformation handling would require a much larger and more balanced evaluation set.

4.8 Error Analysis

Qualitative wins on ambiguity where vanilla RAG lists all candidates.

Table 7. Error analysis summary. Big wins 12% of test set. $N = 75$.

Category	N	% of test set
Big wins ($\Delta F1 > 0.2$)	9	12.0%
Big losses ($\Delta F1 < -0.2$)	5	6.7%
No change ($ \Delta F1 \leq 0.2$)	61	81.3%

Table 7 summarises the distribution of per-example F1 changes between ICCR-RAG and Vanilla RAG. For the majority of examples (61 out of 75, 81.3%), the two systems produce comparable outputs with $|\Delta F1| \leq 0.2$. ICCR-RAG shows notable improvements ($\Delta F1 > 0.2$) on 9 examples (12.0% of the test set) and notable regressions on 5 examples (6.7%). The fact that wins outnumber losses in this analysis is a positive signal, though the overall magnitude of change is limited and the test set is small. Cases of big loss warrant further investigation to understand when ICCR’s re-ranking introduces errors.

Table 8. Top ICCR wins ($\Delta F1 > 0.2$).

Question	Vanilla F1	ICCR F1	$\Delta F1$
When did George Horner die?	0.50	1.00	+0.50
Artist of “Something for Everybody”?	0.57	1.00	+0.43
Composer of Symphony No. 38?	0.60	1.00	+0.40

Table 8 illustrates three representative examples where ICCR-RAG achieves large F1 improvements over the baseline. In each case, vanilla RAG returns a partial or incorrect answer (F1 in the range 0.50–0.60), while ICCR-RAG recovers the fully correct answer (F1=1.00). These examples are factual, unambiguous queries where the correct source document appears to have been ranked higher by ICCR’s support classifier than by cosine similarity alone. It should be noted that these are cherry-picked best cases; the typical improvement across the test set is much smaller, and these examples are not representative of average system behaviour.

4.9 Computational Efficiency

ICCR adds 20ms overhead versus cosine but enables 100× faster inference than multi-agent systems.

Table 9. Computational cost. ICCR adds modest overhead vs. 100 $\times$ speedup vs. multi-agent.

System	Per Query	Memory	LLM Calls	Status
Cosine baseline	30ms	0.5GB	1	Measured
ICCR (ours)	50ms	2.5GB	1	Measured
MADAM-RAG	5–10s	8GB+	5–10	Literature

Table 9 compares the per-query computational cost of the three systems. The cosine baseline processes a query in approximately 30ms using 0.5GB of memory. ICCR adds a modest 20ms overhead, bringing total query time to 50ms while requiring 2.5GB of memory — an increase attributable to loading the MiniLM encoder and BM25 index alongside the classifier. Both systems make a single LLM call per query. In contrast, MADAM-RAG requires 5–10 LLM calls and 5–10 seconds per query according to reported figures in the literature, making it roughly 100–200 $\times$ slower than ICCR. ICCR’s latency and memory footprint are compatible with consumer-grade hardware and single-GPU or CPU-only deployment, which is a practical advantage in resource-constrained settings.

5 Discussion

5.1 Practical Value

1. Misleading@1 reduction (20% $\rightarrow$ 9.3%) directly reduces hallucination risk in the tested setting.
2. CPU-efficient: 50ms/query on 24GB RAM.
3. Interpretable: P_{support} enables routing (escalate high-misinfo queries).
4. No LLM tuning: off-the-shelf Llama3.1-8B.

5.2 Limitations

1. **Statistical power:** $p = 0.58$ (not significant); $N = 75$ limits detection.
2. **Class imbalance:** $N = 5$ misinformation; generalization unclear.
3. **Small dataset:** $N = 75$ test examples.
4. **No-conflict F1=0:** Zero positive examples in test set.
5. **Local BM25:** Ignores global corpus structure.

5.3 When to Use ICCR

Good fit: Production RAG systems with latency constraints.

Poor fit: Research validation requiring statistical significance.

6 Conclusion

ICCR provides a lightweight, preliminary framework for conflict-aware document selection in multi-source RAG. Evaluated on RAMDocs ($N = 75$), it achieves +4.0% EM and reduces misleading top-1 documents by 50% relative to a cosine baseline. It is CPU-deployable (50ms/query) with interpretable confidence signals.

While results are not statistically significant ($p = 0.58$) and are constrained to 75 test examples, the Misleading@1 reduction and computational efficiency suggest ICCR is worth further investigation for production systems. We present these findings as a preliminary study and recommend evaluation on larger, more diverse datasets before drawing broader conclusions.

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